A Rapid+ P+ Extension of Leonetti's Sliding-Jump Theorem

Run fa_banach_001, target arXiv:2201.13059

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Source Problem

The source paper is Paolo Leonetti's arXiv:2201.13059, *Regular matrices of unbounded linear operators*. In the closing remarks, page 24, the source asks whether Theorem 4.1 holds for the larger class of rapid+ P+-ideals, whether this condition characterizes that class, and whether an analogous question holds for Theorem 3.6.

6. Closing remarks

We leave as open questions for the reader to check whether Theorem 4.1 holds for the larger class of rapid$^+$ P^+-ideals $\mathcal{J}$, and whether this condition characterizes the latter class of ideals in the same spirit of [12, Theorem 5.1]. An analogue question could be asked for Theorem 3.6. In addition, it would be interesting to obtain a characterization of the matrix class $(c(X, \mathcal{Z}), c(Y, \mathcal{Z}))$, analogously to Theorem 2.21.

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Definitions and Source Theorem

For an ideal $\mathcal{J}$ on ω, write $\mathcal{J}^+ = \mathcal{P}(\omega) \setminus \mathcal{J}$ and $\mathcal{J}^\star = \{\omega \setminus A : \ A \in \mathcal{J}\}$.

The ideal $\mathcal{J}$ is P+ if every decreasing sequence (S_n) in $\mathcal{J}^+$ has a pseudointersection $S \in \mathcal{J}^+$ such that $S \setminus S_n$ is finite for every n. It is rapid+ if, for every $S \in \mathcal{J}^+$ and every infinite $F \subset \omega$, there is $S' \subset S$ with $S' \in \mathcal{J}^+$ and

$$|S' \cap [0, m]| \leq |F \cap [0, m]| \quad (m \in \omega).$$

Theorem 4.1 of the source says the following when $\mathcal{J}$ is countably generated. Let $A = (A_{n,k})$ be a matrix of linear operators in $\mathcal{L}(X, Y)$ satisfying the source conditions (T1$^\flat$), (T3$^\natural$), and (T6$^\flat$). Then there is a sequence $\mathbf{x} = (x_k)$ with values in the unit sphere S_X such that

$$\mathcal{J}\text{-}\limsup_n \|A_{n,\omega}\| = \mathcal{J}\text{-}\limsup_n \|A_n \mathbf{x}\|.$$

Here $A_{n,E}$ denotes the finite-or-infinite block operator used in the source, and $A_{n,\omega}$ denotes the full row operator.

Idea of the Proof

Leonetti's proof already contains almost all of the argument. It constructs decreasing $\mathcal{J}$-positive "good" sets S_n and then needs a $\mathcal{J}$-positive sequence of row indices s_n with $s_n \in S_n$. Countable generation supplies such a sequence by making the chosen set escape every generator.

For rapid+ P+ ideals there is a replacement selection principle. P+ gives a positive set S almost contained in every S_n. For each n, only finitely many points of S fail to be in S_n. Choose an infinite comparison set F whose nth element lies beyond the first n finite exceptional sets. Rapid+ then thins S to a positive subset whose increasing enumeration is at least as sparse as F. Its nth element is therefore in S_n. With this selected sequence, the original sliding-jump estimates are unchanged.

Lemma 1 (rapid+ P+ diagonal thinning)**.** *Let $\mathcal{J}$ be a rapid+ P+ ideal and let (S_n) be a decreasing sequence in $\mathcal{J}^+$. Then there are a set $R = \{r_0 < r_1 < \cdots\} \in \mathcal{J}^+$ and an integer n_0 such that $r_n \in S_n$ for every $n \geq n_0$.*

Proof. By P+, choose $S \in \mathcal{J}^+$ such that $S \setminus S_n$ is finite for every n. For each n, let

$$b_n = 1 + \max \bigcup_{i \leq n} (S \setminus S_i),$$

with the convention that the maximum of the empty set is 0. Increasing b_n if necessary, choose a strictly increasing infinite set $F = \{f_0 < f_1 < \cdots\}$ with $f_n \geq b_n$ for every n.

By rapid+, there is $R \subset S$ with $R \in \mathcal{J}^+$ and $|R \cap [0, m]| \leq |F \cap [0, m]|$ for all m. Let $R = \{r_0 < r_1 < \cdots\}$. This counting inequality implies $r_n \geq f_n$ for every n: otherwise, if $r_n < f_n$, then at $m = r_n$ the set R has at least $n+1$ points while F has at most n points. Thus $r_n \geq f_n \geq b_n$. Since $r_n \in S$ and r_n is larger than every point of $S \setminus S_n$, we have $r_n \in S_n$ for all n. □

Theorem 1. *The conclusion of Leonetti's Theorem 4.1 remains true if the hypothesis "$\mathcal{J}$ is countably generated" is replaced by "$\mathcal{J}$ is rapid+ and P+".*

Proof. We follow the notation of the source proof. Let

$$\eta_0 = \mathcal{J}\text{-}\limsup_n \|A_{n,\omega}\|.$$

As in the source proof, (T6$^\flat$) implies that finite initial blocks are $\mathcal{J}$-null, and together with (T1$^\flat$) this gives $\eta_0 < \infty$. If $\eta_0 = 0$, any sequence in S_X^ω works. Assume $\eta_0 > 0$.

The source proof defines decreasing sets

$$E_n = \left\{ t \in J_2 : \left| \|A_{t,\omega}\| - \eta_0 \right| \leq \frac{\eta_0}{2^n} \right\} \in \mathcal{J}^+$$

and, after intersecting with finite-block control sets from (T6$^\flat$), obtains a decreasing sequence of good sets $S_n \in \mathcal{J}^+$. In the countably generated proof, one chooses $s_n \in S_n$ escaping the generators. Replace that choice by Lemma 1: pass to $R = \{r_0 < r_1 < \cdots\} \in \mathcal{J}^+$ with $r_n \in S_n$ for all sufficiently large n, and discard the finite initial segment. Relabel the tail as (s_n).

Now choose the increasing integers m_n exactly as in the source proof, using (T3$^\natural$) to ensure

$$\|A_{s_n, \geq m_n}\| \leq \frac{\eta_0}{2^n},$$

and choose vectors $x_k \in S_X$ on the blocks $M_n = (m_{n-1}, m_n]$ so that

$$\left\| \sum_{k \in M_n} A_{s_n,k} x_k \right\| \geq \|A_{s_n, M_n}\| - \frac{\eta_0}{2^n}.$$

Because $s_n \in S_n$, the same estimates as in the source give, for all large n,

$$\|A_{s_n}\mathbf{x}\| \geq \eta_0 \left(1 - \frac{1}{2^{\,n-3}}\right).$$

The selected set $\{s_n : \ n \in \omega\}$ is in $\mathcal{J}^+$, so

$$\mathcal{J}\text{-}\limsup_n \|A_n\mathbf{x}\| \geq \eta_0.$$

The reverse inequality is the elementary row domination inequality used at the start of the source proof:

$$\|A_n\mathbf{x}\| \leq \|A_{n,\omega}\| \quad (n \in \omega).$$

Hence equality holds. □

Verification Notes

Only the selection step in the proof of source Theorem 4.1 changes. All analytic estimates, including the finite-column null estimate, tail control from (T3$^\natural$), and the block-vector choice, are verbatim from Leonetti's proof. The finite initial segment lost in Lemma 1 is irrelevant to $\mathcal{J}$-limsup because finite sets belong to any ideal on ω.

Limitations and Novelty Check

This proves the first extension asked in the closing remarks. It does not prove that Theorem 4.1 characterizes rapid+ P+-ideals, and it does not settle the analogous question for Theorem 3.6 or the matrix class $(c(X, \mathcal{Z}), c(Y, \mathcal{Z}))$.

The run indexes were searched for arXiv:2201.13059 and for terms including "regular matrices", "unbounded linear operators", "rapid+ P+", and "sliding jump". A bounded web search for the source title together with rapid+ P+ and Theorem 4.1 found no exact later answer.

References

1. P. Leonetti, *Regular matrices of unbounded linear operators*, arXiv:2201.13059.
2. B. De Bondt and H. Vernaeve, *Filter-dependent versions of the uniform boundedness principle*, J. Math. Anal. Appl. 495 (2021), Paper No. 124705.